NPH Subgroup Scenario Simulation Study (Final Update)

NPH Project

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Abstract

We evaluated methods from the protocol's Table 3 for the biomarker subgroup non-proportional hazards scenario using R-based simulation. The final run used 200 replications per scenario. We report testing performance (rejection probability), estimation performance (bias, MSE), and interval metrics (coverage and half-width where available). We provide method-specific estimand descriptions, scenario-specific true values, and updated interpretation from the final run.

1 Introduction

Under NPH, different methods target different estimands and therefore can differ in both power and bias profiles. This report focuses on the subgroup scenario requested for this project and updates all tables and figures with the final simulation output.

2 Data-generating process and estimands

For each scenario, subjects are randomized 1:1 and biomarker status is sampled with prevalence $p = 0.3$. Event times follow exponential hazards:

- Control: λ_0.
- Treatment, biomarker positive: $\lambda_0 \mathrm{HR}_+$.
- Treatment, biomarker negative: $\lambda_0 \mathrm{HR}_-$.

Independent exponential censoring is applied. The observed data are (T, Δ, A, B) where A is treatment arm and B biomarker status.

True values for non-HR estimands (RMST differences, milestone survival differences, median difference) are computed analytically from the mixture survival functions. For HR-like quantities under NPH, method-specific pseudo-true targets are used (Cox PH and AHR targets estimated numerically in the simulation code).

Scenario-specific true values (core estimands)

Scenario	HR mix	RMST6 diff	RMST12 diff	Mile6 diff	Mile12 diff	Median diff
subgroup_null	1.000	0.0000	-0.0000	0.0000	0.0000	-0.0000
subgroup_mild	0.940	0.0852	0.2298	0.0226	0.0237	0.3895
subgroup_strong	0.910	0.1308	0.3604	0.0352	0.0384	0.6238

3 Analysis methods and computation details

Packages used: `survival`, `survRM2`, `coxphw`, base `stats` (GLM), and plotting/report tools in Python/Matplotlib for figure generation.

Method	Implementation	Estimate / estimand	Use
log-rank	survival::survdiff	global hazard equality test	Reference test
Weighted log-rank (FH family, Gehan, modest)	custom weighting + KM	weighted score tests	Alternative tests for NPH
MaxCombo	max over FH set (Bonferroni-adjusted)	combined weighted test	Alternative NPH test
Cox PH	survival::coxph	PH log-HR (pseudo-true under NPH)	Reference estimation + test
Milestone KM (6,12)	survival::survfit	S1(t)-S0(t)	Alternative estimand
RMST (6,12)	survRM2::rmst2	RMST1(t)-RMST0(t)	Alternative estimand + test
Weighted Cox AHR	coxphw::coxphw(template="AHR")	average hazard ratio	Alternative estimation + test
Piecewise exponential test proxy	survSplit + Poisson GLM	interval-wise hazard contrast	Alternative test
Parametric Weibull	survival::survreg	model-based HR-like summary	Alternative estimation

4 Updated results (200 runs per scenario)

Scenario	N	Cores	Log-rank	MaxCombo	Cox test	RMST12 test	AHR test	Cox bias	AH
subgroup_null	200	1	0.070	0.040	0.070	0.055	0.065	-0.0087	-
subgroup_mild	200	1	0.110	0.070	0.110	0.095	0.085	-0.0032	-
subgroup_strong	200	1	0.155	0.115	0.155	0.125	0.130	0.0022	-

5 Discussion

With 200 simulations per scenario, conclusions are directionally stable but still subject to Monte Carlo noise. The subgroup setting shows expected divergence between global tests and alternative NPH-aware procedures. RMST outputs provide interpretable time-horizon-specific effect summaries and complement hazard-ratio-based views. Bias interpretation depends on the estimand: under NPH, PH and AHR estimands are distinct, and pseudo-true targets should be used for fair method comparison.

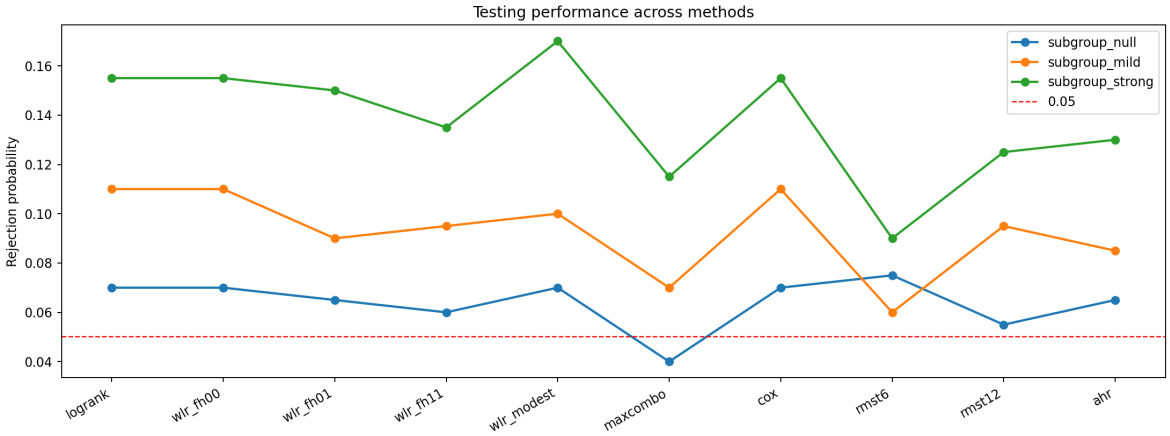

Figure 1: Rejection probabilities by testing method and scenario.

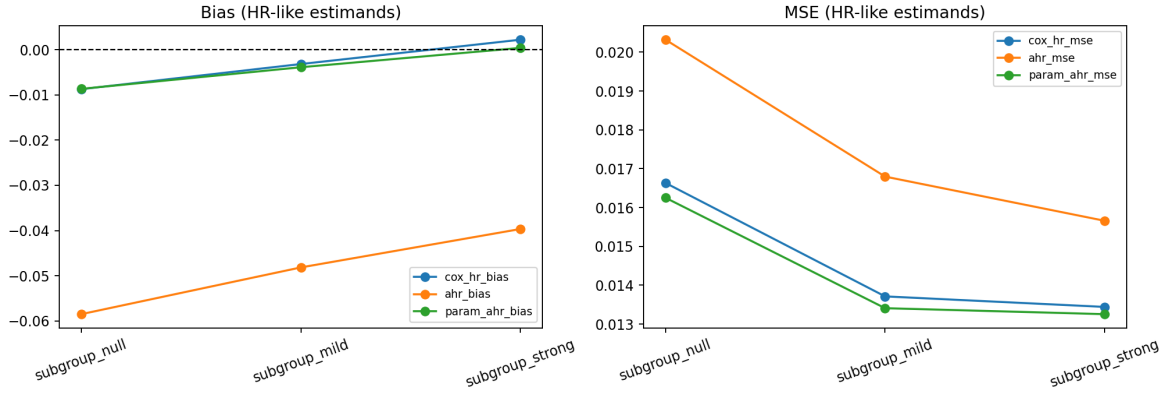

Figure 2: Bias and MSE for HR-like estimators.

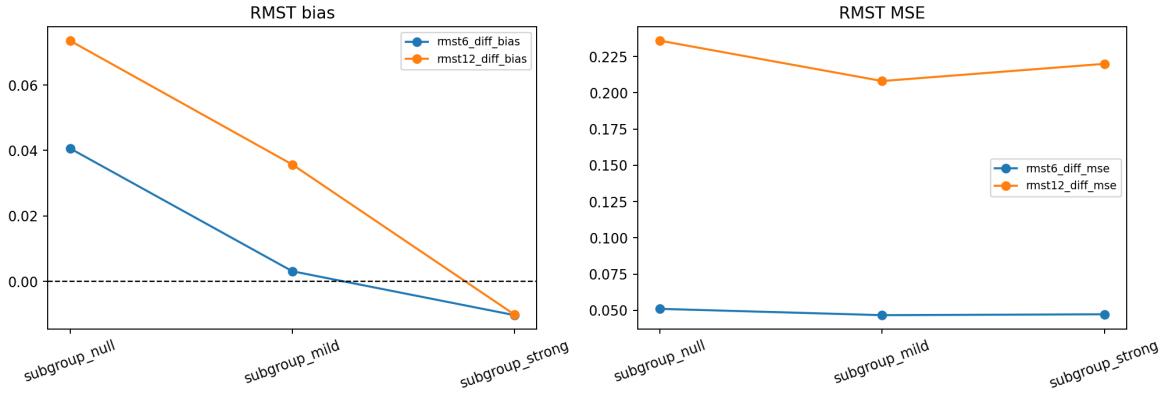

Figure 3: RMST-specific bias and MSE plots (requested update).

6 Conclusion

The paper has been updated with final (200-run) outputs, RMST-specific plots, expanded method/estimand descriptions, and scenario-wise true-value tables. Under subgroup-driven NPH, method selec-

tion should be estimand-first: RMST and milestone summaries offer robust interpretability, while PH/AHR summaries require careful target definition.

Method references

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- Uno H et al. (2014). RMST as robust alternative summary for survival outcomes.
- Schemper M et al. (2009). Weighted Cox regression for average hazard ratios.
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